

DAT Dental Exam

Complete Study Plan

Exam Date: June 24, 2026 | Study Start: May 10, 2026

45 Days · 6 Phases · Full Review Week

Plan Overview

Phase	Dates	Subject	Days
1	May 10 – May 21	Biology	12
2	May 22 – May 31	General Chemistry	10
3	June 1 – June 6	Organic Chemistry	6
4	June 7 – June 11	Perceptual Ability Test (PAT)	5
5	June 12 – June 16	Quantitative Reasoning & Reading Comprehension	5
6	June 17 – June 23	Final Review Week	7
EXAM	June 24, 2026	DAT Dental Admission Test	—

Recommended Daily Schedule

Block	Duration	Activity
Morning	2 hours	New content — read notes, watch videos, take structured notes
Afternoon	2 hours	Practice questions on that day's topic; review all incorrect answers
Evening	30 min	Flashcard review (Anki / paper) — weak spots and formula refresh

Phase 1 — Biology

May 10 – May 21 · 12 Days

Date	Day	Topic / Focus
May 10	1	Cell structure, organelles (nucleus, mitochondria, ER, Golgi, lysosomes), membrane transport (passive, active, osmosis)
May 11	2	Cell division — mitosis (PMAT) & meiosis I/II; crossing over, nondisjunction, ploidy
May 12	3	Molecular biology — DNA structure, semi-conservative replication; transcription, RNA processing; translation & the genetic code
May 13	4	Mendelian genetics — dominant/recessive, Punnett squares, dihybrid crosses, sex-linked traits, codominance, incomplete dominance
May 14	5	Molecular genetics — gene regulation (operons), mutations (point, frameshift), chromosomal abnormalities, cancer biology
May 15	6	Evolution — natural selection, Hardy-Weinberg, speciation, phylogenetics; Ecology — food webs, biomes, population dynamics
May 16	7	REVIEW DAY #1 — Re-read notes Days 1–6; complete 30+ practice questions; make error log
May 17	8	Anatomy I — musculoskeletal (bone types, muscle contraction, sliding filament); circulatory (heart anatomy, cardiac cycle, ABO blood); respiratory (lung mechanics, gas exchange)
May 18	9	Anatomy II — digestive (enzymes, absorption, liver function); excretory (nephron anatomy, filtration, urine formation); nervous system (neuron, action potential, CNS/PNS, reflex arc)
May 19	10	Anatomy III — endocrine (hormones, feedback loops, glands); immune system (innate vs adaptive, B/T cells, antibodies); reproductive systems (gametogenesis, hormonal cycle)
May 20	11	Developmental biology (embryonic layers, organogenesis); Taxonomy (Linnaean classification, 3 domains, 6 kingdoms, key phyla)
May 21	12	FULL BIOLOGY REVIEW — Timed practice section (40 Q / 30 min); review all missed; update formula/term cheat sheet

High-Yield Biology Topics

Content Area	Content Area
• Cell organelle functions	• Cardiac conduction system
• Mitosis vs meiosis stages	• Hormones & their glands
• DNA replication enzymes	• Immune cell types & roles
• Genetic inheritance patterns	• Enzyme kinetics (K_m , V_{max})
• Hardy-Weinberg equilibrium	• Muscle contraction cascade
• Nephron filtration steps	• Embryonic germ layers & derivatives
• Action potential phases	• Taxonomy domains & kingdoms

Phase 2 — General Chemistry

May 22 – May 31 · 10 Days

Date	Day	Topic / Focus
May 22	13	Atomic structure (quantum numbers, electron config, orbital shapes); periodic trends (EN, atomic radius, IE, EA); bonding — ionic, covalent, VSEPR, hybridization
May 23	14	Stoichiometry — molar mass, limiting reagents, percent yield; balancing equations; reaction types (combination, decomposition, single/double displacement, combustion)
May 24	15	Gases — ideal gas law ($PV=nRT$), Dalton's partial pressures, Graham's effusion law, kinetic molecular theory, real gas deviations (van der Waals)
May 25	16	Liquids & solids — IMFs (H-bond, dipole, London), phase diagrams, phase changes; solutions — molarity, molality, colligative properties (BP elevation, FP depression, osmotic pressure)
May 26	17	Thermodynamics — ΔH , ΔS , $\Delta G = \Delta H - T\Delta S$; Hess's law; standard enthalpies of formation; calorimetry ($q = mc\Delta T$)
May 27	18	Chemical equilibrium — K_p & K_c , reaction quotient Q , Le Chatelier's principle; solubility product K_{sp} , common ion effect
May 28	19	Acids & bases — Bronsted-Lowry, K_a/K_b , pH/pOH calculations, strong vs weak acids, buffer systems, Henderson-Hasselbalch, titration curves
May 29	20	Electrochemistry — oxidation states, balancing redox (half-reaction), galvanic vs electrolytic cells, standard cell potential E°_{cell} , Nernst equation, Faraday's law
May 30	21	Kinetics — reaction rates, rate laws, integrated rate laws (0th/1st/2nd order), half-life; Nuclear chemistry — decay types (α , β , γ), half-life calculations, fission vs fusion
May 31	22	FULL GEN CHEM REVIEW — Timed practice section (30 Q / 30 min); review mistakes; update equation sheet

Key Equations to Memorize

Equation	Formula	Use
Ideal Gas Law	$PV = nRT$	Gas problems
Gibbs Free Energy	$\Delta G = \Delta H - T\Delta S$	Spontaneity
Henderson-Hasselbalch	$pH = pK_a + \log\left(\frac{[A^-]}{[HA]}\right)$	Buffer pH
Nernst Equation	$E = E^\circ - \frac{RT}{nF} \ln Q$	Non-standard cell potential
Calorimetry	$q = mc\Delta T$	Heat transfer
First-order half-life	$t_{1/2} = 0.693/k$	Radioactive decay, kinetics

Phase 3 — Organic Chemistry

June 1 – June 6 · 6 Days

Date	Day	Topic / Focus
June 1	23	Nomenclature (IUPAC) for alkanes, alkenes, alkynes; functional groups identification; stereochemistry — chirality, R/S configuration, E/Z alkenes, Fischer projections, optical activity
June 2	24	Substitution reactions — SN1 (carbocation, racemization) vs SN2 (backside attack, inversion); Elimination — E1 vs E2 (Zaitsev's rule); factors: substrate, nucleophile, solvent, temp
June 3	25	Addition reactions — electrophilic addition to alkenes (Markovnikov), HX, H ₂ O, X ₂ , hydroboration; Aromaticity (Huckel's rule, EAS); NAS for benzene
June 4	26	Carbonyl chemistry — aldehydes & ketones (nucleophilic addition, aldol, oxidation/reduction); carboxylic acids & derivatives (esters, amides, anhydrides, acyl substitution); amines (basicity, reactions)
June 5	27	Lab techniques — extraction (acid-base), distillation (simple/fractional), recrystallization, chromatography (TLC, column); spectroscopy overview — IR (functional group ID), NMR (H environments), MS (molecular weight)
June 6	28	FULL ORGO REVIEW — Timed practice section (15 Q / 20 min); draw out every major mechanism from memory; review stereochemistry problems

Reaction Mechanisms Cheat Sheet

Mechanism / Rule	Mechanism / Rule
• SN1 — tertiary substrate, polar protic, carbocation	• Aldol condensation — enolate + carbonyl
• SN2 — primary substrate, polar aprotic, backside attack	• Ester hydrolysis — acid or base catalyzed
• E1 — tertiary, high temp, weak base	• Grignard reaction — carbon nucleophile
• E2 — strong base, anti-periplanar geometry	• Oxidation states — primary → aldehyde → acid
• Electrophilic addition — Markovnikov rule	• Reduction — NaBH ₄ (ketones/aldehydes) vs LiAlH ₄ (all carbonyls)
• EAS — activating vs deactivating groups	• Diels-Alder — diene + dienophile → cyclohexene

Phase 4 — Perceptual Ability Test (PAT)

June 7 – June 11 · 5 Days

Date	Day	Topic / Focus
June 7	29	Aperture Passing (Keyhole) — visualize 3D objects passing through 2D cutouts; practice 60 items timed; learn elimination strategy
June 8	30	View Recognition — orthographic projection; match top / front / end views to 3D object; practice drawing views from isometric diagrams

June 9	31	Angle Discrimination — rank angles from smallest to largest; Paper Folding — track fold/punch positions mentally; 50 timed items each
June 10	32	Cube Counting — count hidden cubes, track exposed faces; 3D Form Development (Pattern Folding) — fold flat nets into 3D boxes; identify correct assembled shape
June 11	33	FULL PAT PRACTICE TEST — all 6 PAT subtypes timed (90 Q / 60 min); score & analyze weakest subtype; drill that subtype with 30 extra items

PAT Strategy Tips

- Aperture: check overall shape silhouette first — eliminate non-matches before testing orientation.
- View Recognition: always identify the top view first; it eliminates the most options.
- Angle Discrimination: use a reference angle (90°) to bucket choices, then rank within buckets.
- Paper Folding: trace one fold at a time; mark the punch dot relative to the fold line.
- Cube Counting: count systematically by layer (bottom → top); mark cubes as you go.
- Pattern Folding: identify which face must be opposite which, and eliminate impossible answers.
- Pacing: ~40 sec per item — skip and flag if stuck; return at the end.

Phase 5 — Quantitative Reasoning & Reading Comprehension

June 12 – June 16 · 5 Days

Date	Day	Topic / Focus
June 12	34	QR — Algebra: solving equations, inequalities, systems, quadratics; Word problems (rate, distance, work, mixture); Unit conversions & proportions
June 13	35	QR — Probability & statistics (mean, median, mode, SD, combinations/permutations); Geometry (area, volume, Pythagorean theorem); Trigonometry basics (SOH-CAH-TOA)
June 14	36	RC — Reading strategy: search-and-destroy vs read-all-first; identifying tone, main idea, inference questions; complete 2 timed passages with full review
June 15	37	RC — 3 full timed passages (natural sciences format); review all incorrect; note recurring trap answer types
June 16	38	FULL-LENGTH TIMED PRACTICE EXAM — all DAT sections under test conditions; log score by section; identify top-3 weak areas for final review week

QR Topics at a Glance

Quantitative Reasoning Topic	Quantitative Reasoning Topic
• Algebra — linear equations, systems, inequalities	• Statistics — mean, median, mode, range, SD
• Quadratic equations & factoring	• Geometry — area & perimeter of 2D shapes
• Ratios, proportions, percentages	• Volume & surface area of 3D solids
• Rate, distance, work problems	• Coordinate geometry — slope, midpoint, distance
• Probability — independent & conditional events	• Trigonometry — SOH-CAH-TOA, unit circle basics

• Combinations & permutations (nCr, nPr)

• Logarithms & exponent rules

Phase 6 — Final Review Week

June 17 — June 23 · 7 Days

Date	Day	Topic / Focus
June 17	39	BIOLOGY REVIEW — High-yield only: cell division, genetics, organ systems, homeostasis. Re-read Feralis notes. 20 mixed practice questions.
June 18	40	GEN CHEM REVIEW — High-yield: acid/base, equilibrium, electrochemistry, thermodynamics. Run through equation sheet. 20 mixed practice questions.
June 19	41	ORGO REVIEW — Mechanisms cheat sheet from memory; SN1/SN2/E1/E2 decision tree; carbonyl reactions. 30-min PAT warm-up session.
June 20	42	QR & RC REVIEW — Drill weakest QR topic from practice exam; complete 1 timed RC passage; review answer strategies for both sections.
June 21	43	FULL-LENGTH PRACTICE TEST #2 — timed, exam-day conditions; simulate every break and transition exactly as on test day.
June 22	44	REVIEW TEST #2 RESULTS — focus only on weak areas flagged by test; no new material; consolidate formula sheet to one page; prepare test-day checklist.
June 23	45	LIGHT DAY — review formula sheet (30 min max); visualize exam-day routine; pack materials (ID, snacks, water); eat well; sleep by 10 PM.

EXAM DAY — June 24, 2026

- Eat a real breakfast — avoid heavy or unfamiliar foods.
- Arrive at the testing center 30 minutes early with valid photo ID.
- Section order: Survey of Natural Sciences → PAT → Reading Comprehension → Quantitative Reasoning.
- Use scratch paper heavily during PAT and QR; show all work.
- Flag uncertain questions and return — never leave a question blank (no penalty).
- Trust the 45 days of preparation. Stay calm and pace yourself.

Recommended Study Resources

Resource	Best Used For	Priority
Chad's Videos	Gen Chem & Orgo concept videos — best explanations available	★★★★★
Feralis Biology Notes	Highest-yield biology outline; covers ~95% of DAT bio content	★★★★★
Crack DAT PAT	Daily PAT drills; most realistic PAT practice online	★★★★★
DATBooster / Bootcamp	Full-length practice tests; question banks for all sections	★★★★★

Anki (DAT deck)	Spaced-repetition flashcards; bio + chem vocabulary	★★★★■
Kaplan DAT Prep	Content review book; good for structured reading	★★★██
Khan Academy	Free backup for gen chem and math concept review	★★★██

Target Scores (Competitive Range)

Section	Competitive Score	Top-Tier Score
Biology	18–19	20+
General Chemistry	18–19	20+
Organic Chemistry	18–19	20+
Perceptual Ability Test	18–19	20+
Reading Comprehension	18–19	20+
Quantitative Reasoning	17–18	20+
Academic Average (AA)	19	21+

Good luck — 45 days of consistent, focused work will get you there. You've got this.